Homework HW#05 announced on 6 Dec 2024.

True (select A in the homework system) or false (B).

Consider the following forwarding table.

Destination IP Address Range	**Link Interface**
11100000 00000000 00000000 00000000 through 11100000 00111111 11111111 11111111	0
11100000 01000000 00000000 00000000 through 11100000 01000000 11111111 11111111	1
11100000 01000001 00000000 00000000 through 11100001 01111111 11111111 11111111	2
otherwise	3

Q1. A datagram whose destination IP is 11001000 10010101 11010001 00010111, and this datagram will be forwarded to interface 3?

Q2. 11100001 01111111 11001011 10111111 will go to interface 2?

Q3. 11100001 11000000 00010001 01110111 will go to interface 3?

Q4. Give an IP 192.168.51.1 and its netmask is /23. The subnet ID of this LAN is 11000000 10101000 00110010 00000000.

Q5. Give an IP 192.168.51.1 and its netmask is /24. The subnet ID of this LAN is 11000000 10101000 00110011 00000000.

Q6. A /27 subnet has a mask 255.255.255.224 and it contains 32 addresses.

Q7. Give a host IP address 192.168.1.123 in a subnet with a netmask 255.255.255.240. The last address of this subnet is 192.168.1.127.

Q8. Given a 172.16.32.1 IP in a /20 subnet, the subnet’s last IP address is 172.16.47.255.